

Q1.

The random variable X has an exponential distribution with mean μ . The cumulative distribution function of X for $x \geq 0$ is given by

$$F(x) = 1 - e^{-\frac{1}{\mu}x}$$

- (a) Find an exact expression for the interquartile range of X in terms of μ .

(5)

- (b) Prove, by integration, that $E(X^2) = 2\mu^2$.

(4)

- (c) Show that the standard deviation of X is less than the interquartile range of X .

(3)

(Total 12 marks)

Q2.

The random variable X may be modelled by an exponential distribution with probability density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that, for $x \geq 0$, the cumulative distribution function is given by

$$F(x) = 1 - e^{-\lambda x}$$

(2)

- (b) For an exponential distribution with $\lambda = 0.5$, find values for the two probabilities missing from the following table, giving your answers to four decimal places.

Interval	0 – 1	1 – 2	2 – 3	3 – 4	4 – 5	≥ 5
Probability	0.3935	0.2387	0.1447	0.0878		

(2)

(Total 4 marks)

Q3.

The continuous random variable X is modelled by an exponential distribution with probability density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- (a) Write down the cumulative distribution function, $F(x)$.

(2)

- (b) Show that the **exact** value of the interquartile range of X is given by $\frac{1}{\lambda} \ln 3$. (5)
- (c) (i) Use integration to prove that $E(X^2) = \frac{2}{\lambda^2}$. (4)
- (ii) Hence, given that $E(X) = \frac{1}{\lambda}$, show that $\text{Var}(X) = \frac{1}{\lambda^2}$. (1)
- (d) (i) Find the **exact** value of λ for which the value of the interquartile range of X is four times the value of the variance of X . (2)
- (ii) Describe what happens to the interquartile range of X as λ increases without bound. (1)
- (Total 15 marks)

Q4.

The lifetimes of electrical components follow an exponential distribution with mean 200 hours.

- (a) Calculate the probability that the lifetime of a randomly selected component is:
- (i) less than 120 hours; (2)
- (ii) more than 160 hours; (2)
- (iii) less than 160 hours, given that it has lasted more than 120 hours. (3)
- (b) Determine the median lifetime of these electrical components. (3)
- (Total 10 marks)

Q5.

- (a) A continuous random variable X has probability density function

$$f(x) = \lambda e^{-\lambda x} \text{ for } x \geq 0$$

Prove that the mean value of X is $\frac{1}{\lambda}$.

- (b) The lifetime of a component in a machine is T hours, where T has probability

density function

$$f(t) = \frac{1}{\alpha} e^{-\frac{t}{\alpha}} \text{ for } t \geq 0$$

The mean lifetime of these components is known to be 62.5 hours.

- (i) Find the value of $\frac{1}{\alpha}$. (2)
- (ii) Calculate the probability that a component will last for at least 80 hours. (4)
- (iii) Given that a component has lasted for 80 hours, find the probability that it will last for a further 20 hours. (3)
- (Total 13 marks)

Q6.

- (a) The time, in years, that a taxi driver keeps his taxi, before replacing it with a new one, can be modelled by an exponential distribution with parameter 0.2.

Find the probability that he keeps his taxi:

- (i) for less than 2 years; (2)
- (ii) for more than 3 years. (2)

- (b) The continuous random variable X has an exponential distribution with probability density function $f(x)$, where

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$$f(x) = \lambda e^{-\lambda x} \text{ for } x \geq 0$$

Use integration to find:

- (i) $E(X)$; (4)
- (ii) the median value of X . (3)
- (c) Breakdowns occur on a stretch of road at a mean rate of 0.3 per day. The number of breakdowns follows a Poisson distribution.

Find, in hours:

- (i) the mean time between breakdowns; (1)
- (ii) the median time between breakdowns.

(2)
(Total 14 marks)

Q7.

The random variable T has an exponential distribution with mean 5.

- (a) (i) Specify fully the probability density function, $f(t)$, for T .

(2)

- (ii) Show that, for $t \geq 0$, T has distribution function

$$F(t) = 1 - e^{-0.2t}$$

(2)

- (iii) Find the probability that a randomly chosen value of T lies between 2 and 8.

(3)

- (b) A roadside snack bar opens at 7.30 am each day. There is a probability of 0.15 that at least one customer is already waiting when the snack bar opens. Otherwise the time, T minutes, until the first customer arrives can be modelled by an exponential distribution with mean 5.

- (i) On a particular day, there are no customers waiting when the snack bar opens.

Find the probability that no customers have arrived by 7.40 am.

(2)

- (ii) For a randomly selected day, find the probability that the first customer arrives before 7.35 am.

(3)

(Total 12 marks)

Q8.

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The pupils in a school are affected by an outbreak of flu. Any pupil who catches flu is ill for at least three full days. After this three-day period there is a **further** time, T days, until the pupil has fully recovered.

The random variable, T , is exponentially distributed with the following probability density function.

$$f(t) = \begin{cases} \frac{1}{2} e^{-\frac{t}{2}} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

- (a) Show that, for $t \geq 0$, T has distribution function

$$F(t) = 1 - e^{-\frac{t}{2}}.$$

(2)

- (b) (i) Janice has been ill with flu for exactly three days. Calculate the probability

that she will have fully recovered in a **further** three days.

(2)

- (ii) Kevin has just caught flu. Calculate the probability that he will **not** have fully recovered in four days' time.

(3)

- (iii) Lucy has been ill with flu for exactly four days. Calculate the probability that she is ill with flu for a **total** of less than six days.

(4)

- (c) The random variable, X , denotes the **total time**, in days, for which a pupil is ill with flu. Write down values for the mean and standard deviation of X .

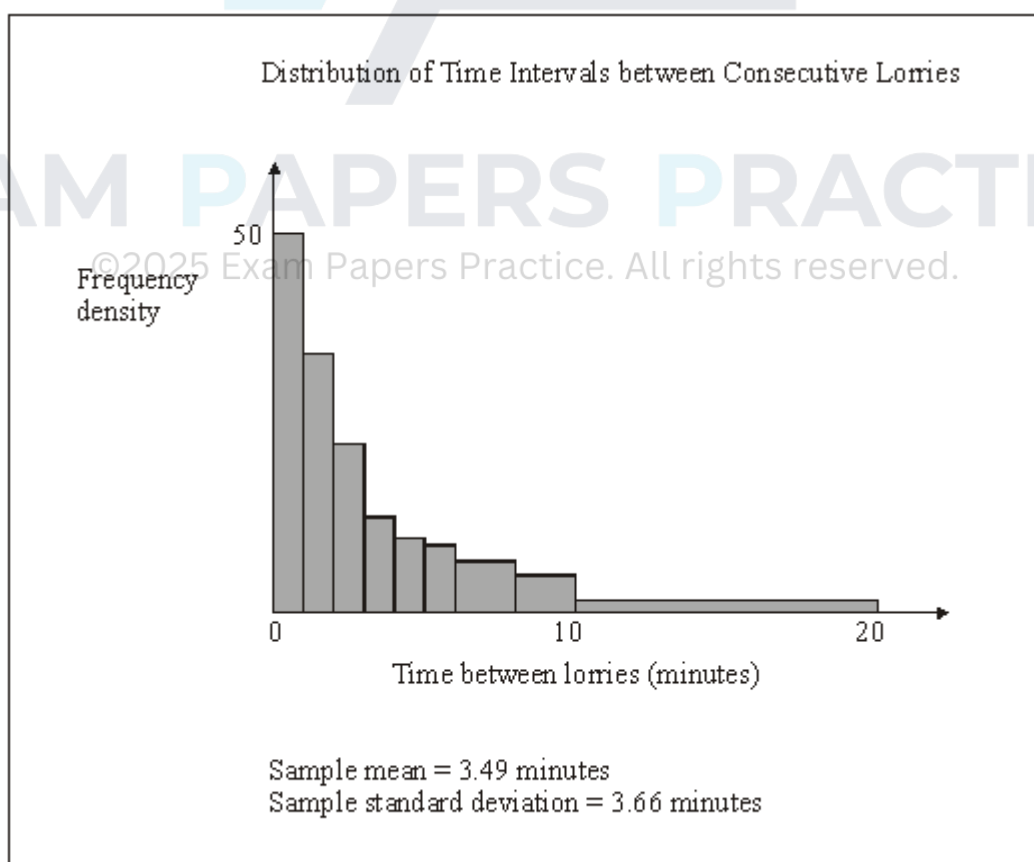
(3)

(Total 14 marks)

Q9.

The parents of children at a village primary school are concerned about the frequency of lorries passing the school. The Parents' Association organises a survey. Over a three-week period, volunteers stand outside the school, at randomly selected times during the school day, and record the time interval between consecutive lorries driving past in either direction.

When the survey is complete, the secretary of The Parents' Association summarises the data by drawing a histogram and calculating the mean and standard deviation of the time intervals between consecutive lorries. The results are shown below.



- (a) Using the above summary of data, state reasons why it might be appropriate to use

an exponential distribution to model the time intervals between consecutive lorries.

(2)

- (b) The distribution of the time interval, T minutes, between consecutive lorries is to be modelled as an exponential distribution with distribution function:

$$P(T \leq t) = F(t) = \begin{cases} 1 - e^{-0.3t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

- (i) Find the mean value of T .

(1)

- (ii) Find the probability that the time interval between consecutive lorries is less than one minute.

(2)

- (iii) A school crossing patrol lady helps children to cross the road outside the school. One morning she steps out into the road to stop the traffic exactly one minute after a lorry has passed. She then holds up the traffic for exactly 45 seconds. Find the probability that no lorries are waiting when she allows the traffic to move.

(3)

- (iv) Calculate the median time interval between consecutive lorries.

(3)

(Total 11 marks)

Q10.

The times, T minutes, between successive telephone calls to a small business can be assumed to be exponentially distributed with mean 10.

- (a) (i) Write down, in full, the probability density function of T .

(2)

- (ii) Using integration, obtain an expression, valid for $t \geq 0$, for the distribution function of T .

(3)

- (b) A call is received at 11.00 am.

Find the probability that the next call is received:

- (i) before 11.15 am;

(2)

- (ii) between 11.10 am and 11.20 am.

(2)

(Total 9 marks)

Q11.

John travels to work by bus and he arrives at the bus stop at exactly 8 am every morning.

From past experience, he knows that the next bus arrives at X minutes past 8, where X is an exponentially distributed random variable with mean 10.

- (a) Write down the probability density function of X . Deduce that the distribution function of X is given, for $x \geq 0$, by

$$F(x) = 1 - e^{-\frac{x}{10}} \quad (4)$$

- (b) Calculate the probability that no bus has arrived by 8.15 am.

(2)

- (c) Given that no bus has arrived by 8.15 am, calculate the probability that the next bus arrives before 8.20 am.

(3)

(Total 9 marks)

Q12.

The lifetime, T hours, of a certain electrical component is exponentially distributed with mean 800.

- (a) (i) State, in full, the probability density function of T .

(2)

- (ii) Write down the numerical value of the standard deviation of T .

(1)

- (iii) Calculate the probability that one of these components, chosen at random, is still working after 1000 hours.

(3)

- (iv) Given that one of these components is still working after 1000 hours, calculate the probability that it is still working after a further 1000 hours.

(3)

- (b) A piece of equipment contains four of these components. The design is such that it is still functional when at least two of the four components are working. Calculate the probability that the piece of equipment is still functional after 1000 hours.

(4)

(Total 13 marks)

Q13.

Trains stop at a particular station for a minimum of 1 minute.

The time, T minutes, **in excess of 1 minute**, that trains stop at the station may be modelled by an exponential distribution with mean 2.

- (a) State the probability that a train stops at the station for at least 1 minute.

(1)

- (b) Calculate the probability that a train stops at the station for more than 5 minutes.

(3)

- (c) A train has been stopped at the station for 3 minutes.

Calculate the probability that it stops at the station for **a total of** less than 5 minutes.

(4)

- (d) A random sample of five trains each stop at the station for more than 5 minutes.

Indicate why this casts doubt on the above model.

(2)

(Total 10 marks)

Q14.

In a particular county, the distance between successive potholes on country roads may be modelled by an exponential distribution with mean 125 metres.

Determine the probability that the distance between successive potholes is:

- (a) less than 100 metres;

(2)

- (b) between 100 metres and 300 metres.

(3)

(Total 5 marks)

Q15.

Between 8 am and 6 pm, the time, T minutes, between successive arrivals of vehicles for fuel at a village garage may be assumed to have an exponential distribution with a mean of 8.

- (a) Write down the numerical value for the standard deviation of T .

(1)

- (b) Calculate the probability that the time between successive arrivals of vehicles for fuel at the garage is between 5 minutes and 15 minutes.

(3)

- (c) Between 6 pm and 9 pm, the time, S minutes, between successive arrivals of vehicles for fuel at the garage may be assumed to be independent of T and to have an exponential distribution with a mean of 15.

Calculate the probability that no vehicles arrive for fuel at the garage between 5.45 pm and 6.15 pm.

(5)

(Total 9 marks)

Q16.

The time, D days, between successive accidents at a factory can be modelled by an exponential distribution with mean 16.

- (a) Write down the numerical value for the standard deviation of D .

(1)

- (b) Calculate the probability that the time between successive accidents at the factory is

- (i) more than 20 days;
- (ii) between 20 and 30 days.

(5)

- (c) Given that there are no accidents during a 20-day period, determine the probability that there are no accidents during the next 20 days. Justify your answer.

(3)

- (d) Given that the factory is open five days each week, specify the distribution of the **weekly number** of accidents at the factory.

(2)

(Total 11 marks)

Q17.

The lifetime, X **thousand** hours, of a component in a photocopier can be modelled by an exponential distribution with

$$P(X \leq x) = 1 - e^{-\frac{x}{6}}, \quad x \geq 0.$$

Determine the probability that the lifetime of a component is:

- (a) less than 3000 hours;
- (b) between 5000 hours and 10 000 hours.

(2)

(3)

(Total 5 marks)

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